



PROJECT PLAN

TNSDC Virtual Internship Program

IBM Agentic AI Track

1. Program Overview

Program Name	TNSDC Virtual Internship Program on IBM Agentic AI
Delivery Partner	Adroit Technologies Innovative Solutions Pvt. Ltd. in association with IBM
Mode	Online / Virtual (Live Instructor-Led)
Duration	5 Days
Dates	31 August 2026 (Monday) – 4 September 2026 (Friday)
Daily Timing	9:30 AM – 4:00 PM
Track	Agentic AI (IBM)
LMS	Adroit ProLearn
Students Count	5279

2. Program Highlights

- Access to a global IBM certification examination for every participating student
- Hands-on exposure to industry-oriented Agentic AI tools — LangChain, LangGraph, RAG and MCP
- Complete hands-on sessions — every concept reinforced with a guided lab or exercise
- Students build a real, working Agentic AI project from Day 1 through Day 5
- Live, instructor-led delivery.
- Final project submitted with full documentation and a GitHub repository link
- LMS access to students for tracking progress (Submissions, Quiz, Digital contents, Feedback)

3. Program Objective

To equip students with practical, hands-on capability to design and build an Agentic AI application — from concept to a fully working, demoable project — using industry-standard tools (LLMs, LangChain, LangGraph, RAG, MCP), while preparing them for IBM's global certification exam.

4. Program Outcome

By the end of the program, students will be able to:

- Explain agentic AI fundamentals and core agent architecture (model, tools, memory)
- Build and test an AI agent with tool-calling using LangChain
- Design a multi-step agent workflow using LangGraph (nodes, edges, state, memory, error handling)
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- Build a Retrieval-Augmented Generation (RAG) pipeline and extend it into Agentic RAG with MCP integration
- Independently build, test, document and present a complete, working Agentic AI project with a GitHub repository

5. Day-Wise Plan

Day	Focus	5-Hour Session Plan	Project Output
Day 1	Agentic AI & Project Foundation	Agentic AI fundamentals & use case selection; project architecture & requirements; Python/LLM setup + Ollama; build first LLM-powered component	Project idea finalized + basic AI application
Day 2	Building the AI Agent	Agent, Model, Tools & Memory; tool calling with LangChain; build agent with multiple tools; testing & debugging	Working AI Agent with tool calling
Day 3	LangGraph & Agent Workflow	LangGraph concepts; nodes, edges & state; build multi-step agent workflow; add memory & error handling	Functional multi-step Agentic workflow
Day 4	Agentic RAG + MCP	Traditional RAG vs Agentic RAG; document processing & retrieval; build Agentic RAG module; MCP integration (intro + hands-on)	Agent with knowledge base + external tool/MCP capability
Day 5	End-to-End Project & Demo	Integrate all components; complete project development; testing, optimization & documentation; project presentation & evaluation	Final working Agentic AI project

6. Use Cases (Students select 1 on Day 1)

Each student selects one use case on Day 1 and develops it into their final project across the 5 days.

#	Use Case	What Students Build	Key Agent Capabilities
1	AI Student Support Assistant	Answers college-related questions from regulations, syllabus, FAQs and notices	RAG + Tools + Memory
2	AI HR Recruitment Assistant	Screens resumes, matches candidates with job descriptions and generates interview questions	Agent + Tools + RAG
3	AI E-Commerce Customer Support Agent	Handles product queries, order-status requests, returns and recommendations	Tool Calling + Memory
4	AI IT Helpdesk Agent	Diagnoses common technical issues and recommends troubleshooting steps using a knowledge base	Agent + RAG + Tools
5	AI Learning & Study Assistant	Creates learning plans, answers questions from course materials and generates quizzes	RAG + Memory + Tools



7. Daily Time Structure (9:30 AM – 4:00 PM)

Time Slot	Activity
9:30 AM – 11:00 AM	Live concept session
11:00 AM – 11:15 AM	Break
11:15 AM – 1:00 PM	Hands-on lab / guided practice
1:00 PM – 1:45 PM	Lunch break
1:45 PM – 3:15 PM	Applied exercise / project work
3:15 PM – 4:00 PM	Q&A, doubt-clearing, day wrap-up

8. LMS — Adroit ProLearn

- All students enrolled on Adroit ProLearn
- Day 3: Quiz conducted on the LMS
- Day 5: Final project document and GitHub repository link submitted on the LMS

9. Attendance

- Daily QR code shared during the live call — once in the morning session, once in the afternoon session
- Students scan and register attendance both times, each day

10. Feedback

- Collected on Day 4 via Adroit ProLearn

11. Certification

- IBM CEP (Certification Exam Prep) access shared on Day 1 to each student's registered email ID

13. Pre-Program Checklist

- Share program posters with the participating colleges
- Connect with the college SPOC to confirm the program
- Send confirmation mails to SPOCs and mentors
- Share individual meeting links with students via email
- Provision IBM CEP and Adroit ProLearn access for all students

14. Post-Program Checklist

- Project report along with attendance record
- Consolidated project GitHub links
- Student feedback
- IBM CEP certification completion list